SOUTHEAST UNIVERSITY

Assignment on: Simple LAN Design and Configure with Class B net ID.

Course Title: ETE 282.8

Course Code:

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Experiment Name: Simple LAN Design and Configure with Class B Net ID.

Introduction:

- Computer networking serves as the vital backbone of modern communication systems. A local area network (LAN) provisions connectivity among multiple end-devices within a contiend geographical area, such as a localized classroom, corporate office, buildings or university campus. In this lab session, we architected and configured a localized network topology utillizing a class B IP addressing scheme within the Cisco Packet Tracer simulation environment. The primary purpose of this experiment was to establish a foundational understanding of IP addressing mechanics, end-device configuration, and intra-network data communication.

Objectives:

- To design and simulate a functional Local Area Network (LAN) using Cisco Packet Tracer.
- To program network devices using a private class B IP addressing standard.

- To comprehend the logical structure of IP addressing and its role in node-to-node network communication.
- To validate end-to-end network connectivity utilizing the Packet Internet Groper (Ping) utility.

Theoretical Background:

1. Local area network (LAN):
   - A local area network is an infrastructure designed to interconnect computing devices within a localized physical scope. The primary utility of a LAN is to facilitate high-speed data transfer, allow nodes to communicate efficiently and enable the seamless sharing centralized resources such as application servers and networking hardware.

2. Internet Protocol version 4 (IPV4) Addressing:
   - An IPV4 address is a logical 32-bit numerical identifier assigned to each participating device in a network, ensuring unique identification. It is conventionally represented in a dotted-decimal format.
   - Structure: The 32 bits are divided into 4 distinct octets with each octet capable of representing decimal values from 0 to 255.

- Components: Every IP address is bifurcated into two logical segments: the Network ID (identifying the specific network segment) and the Host ID (identifying the specific device on that segment).

3. Classful Addressing and Class B Specifications:

In the traditional Classful addressing architecture, the IP space is categorized into five distinct classes (A, B, C, D and E) determined by the leading bits of the first octet.

- Class B Characteristics: Class B networks are designed for medium to large scale environments. The first octet value ranges strictly from 128 - 191.

- Subnet mask: The default subnet mask for Class B is 255.255.0.0. This allocates 16 bits for the network portion and 16 bits for the host portion, theoretically supporting up to 65,534 usable hosts per network.

4. ICMP and Connectivity Verification:

The Ping command operates using the Internet Control Message Protocol (ICMP). When a node attempts to verify reachability to a target, it transmits an ICMP Echo Request. If the target is operational and reachable, it responds with an ICMP Echo Reply.

A successful reply definitively confirms Layer 3 connectivity.

Network Topology and Hardware Requirement:

The simulated topology follows an expanded star-network model centered around a single Layer 2 switch.

- Hardware used: 1x Cisco Router, 1x Cisco 2960 Switch, 5x Desktop PCs, 2x laptops and 1x server.
- Transmission media: Devices are interconnected via copper straight-through cabling using FastEthernet (Fa) interfaces.

IP Addressing Table:

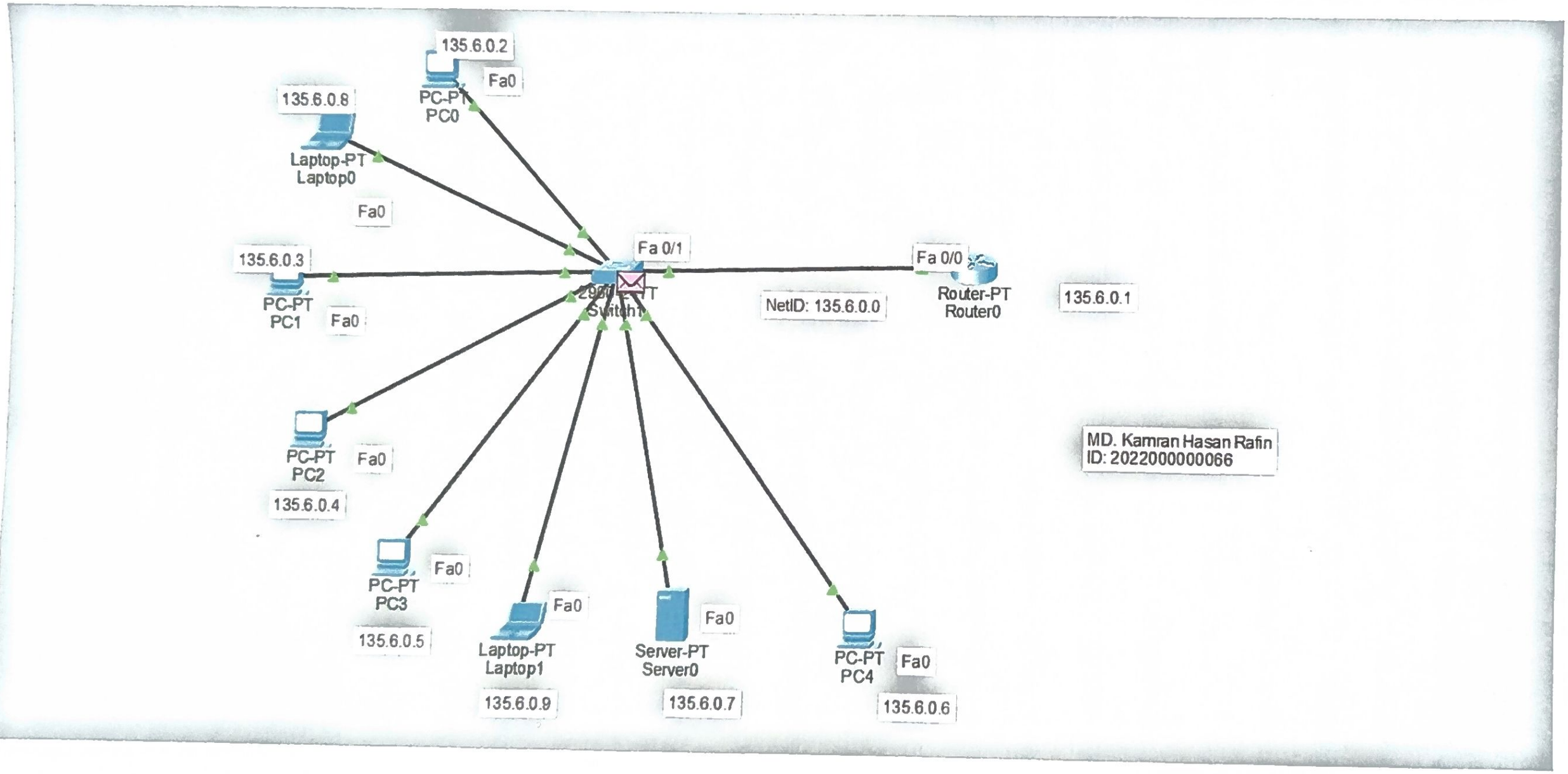
The topology utilizes the 135.6.0.0 Class B network space. Because all end-devices share the same network ID, they reside within the same broadcast domain.

Device Hostname	Interface	Assigned IP Address	Subnet mask	Default gateway
Router 0	Fa 0/0	135.6.0.1	255.255.0.0	N/A
PC 0	Fa0	135.6.0.2	255.255.0.0	135.6.0.1
PC1	Fa0	135.6.0.3	255.255.0.0	135.6.0.1
PC2	Fa0	135.6.0.4	255.255.0.0	135.6.0.1
PC3	Fa0	135.6.0.5	255.255.0.0	135.6.0.1
PC4	Fa0	135.6.0.6	255.255.0.0	135.6.0.1
Server 0	Fa0	135.6.0.7	255.255.0.0	135.6.0.1
Laptop 0	Fa0	135.6.0.8	255.255.0.0	135.6.0.1
Laptop 1	Fa0	135.6.0.9	255.255.0.0	135.6.0.1

Implementation and Configuration steps:

1. Router Interface configuration:

- The router acts as the default gateway for the network. It was configured via the command Line Interface (CLI):
  - Transition to Privileged EXEC mode: `Enable`
  - Enter global configuration mode: `Configure terminal`
  - Select the target fastEthernet interface: `interface Fa0/0`
  - Provision the IP address: IP address `135.6.0.1 255.255.0.0`
  - Bring the interface to an 'up' state: `no shutdown`


135.6.0.2
Fa0
PC-PT
PC0
135.6.0.8
Laptop-PT
Laptop0
Fa0
135.6.0.3
PC-PT
PC1
Fa0
Fa 0/1
Switch1
NetID: 135.6.0.0
Fa 0/0
Router-PT
Router0
135.6.0.1
PC-PT
PC2
Fa0
135.6.0.4
PC-PT
PC3
Fa0
135.6.0.5
Laptop-PT
Laptop1
Fa0
135.6.0.9
Server-PT
Server0
Fa0
135.6.0.7
PC-PT
PC4
Fa0
135.6.0.6
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2. End-Device (PC/Laptop) Configuration:

- Access the device interface and navigate to the Desktop tab, then IP configuration.
- Input the static IP address corresponding to the device (135.6.0.2) and the (255.255.0.0) subnet mask.
- Define the default gateway as 135.6.0.1 the router's IP to allow future external routing capabilities.

3. Server Provisioning:

- Navigate to Desktop -> IP configuration.
- Assign the static IP 135.6.0.7 and the appropite subnet mask and gateway parameters.
- Ensure HTTP services are enabled to allow the server to provide centralized application services ~~access~~ across the LAN.

Operational working Principle:

Data transmission within this LAN relies on Layer 2 switching mechanics.

- Source Evaluation: The originating end-device evaluates the destination IP against its own subnet mask to determine if the target is local.

- Layer 2 Forwarding: Because all communicating nodes share the same network segment, the router is bypassed for Internet traffic. Instead, switch 1 forwards Ethernet frames dynamically based on the target MAC address.

- ICMP Resolution: when a PC executes a Ping to server0, an ICMP echo request is encapsulated and transmitted. server0 receives the payload, processes the request and dispatches an ICMP echo reply back to the originating PC.

Network Advantages:

- Simplified Architecture: The star topology ensures a highly logical and easy to configure design.

- Scalability: By utilizing a Class B subnet mask, the network inherently supports a vast magnitude of host devices, making it highly suitable for medium to large organizational deployments.

- High Efficiency: Intra-network traffic is switched entirely at layer 2 without routing overhead, ensuring efficient, high-speed data communication.

Result and Conclusion:

- The localized topology was successfully provisioned using a Class B private addressing scheme. The router's gateway interface was successfully activated, and ICMP connectivity tests validated flawless, and bidirectional communication among all simulated nodes. The LAN operated optimally with zero measured packet loss.
- In conclusion, this lab experiment provided critical hands on application of networking fundamentals. We successfully bridged theoritical concepts - such as the structural classification of IP addresses, subnet masking and network/Host ID differentiation with practical implementation techniques. The exercise solidified our capability to manually assign IP parameters to a larger pool of devices and configure gateway routing interfaces and troubleshoot connectivity utilizing ICMP diagnostic tools.